MATH 180 Strategies of Problem Solving

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Outline

1. SoPS Lecture 19
   - Functions
   - Cardinality
   - Wrapping up

Review

Last time

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- Sets, Subsets, Intersection, Union, Complements

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- Sets, Subsets, Intersection, Union, Complements
- Cardinality of finite sets, power set

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Last time

- Sets, Subsets, Intersection, Union, Complements
- Cardinality of finite sets, power set
- Universal quantifiers, Russell's paradox

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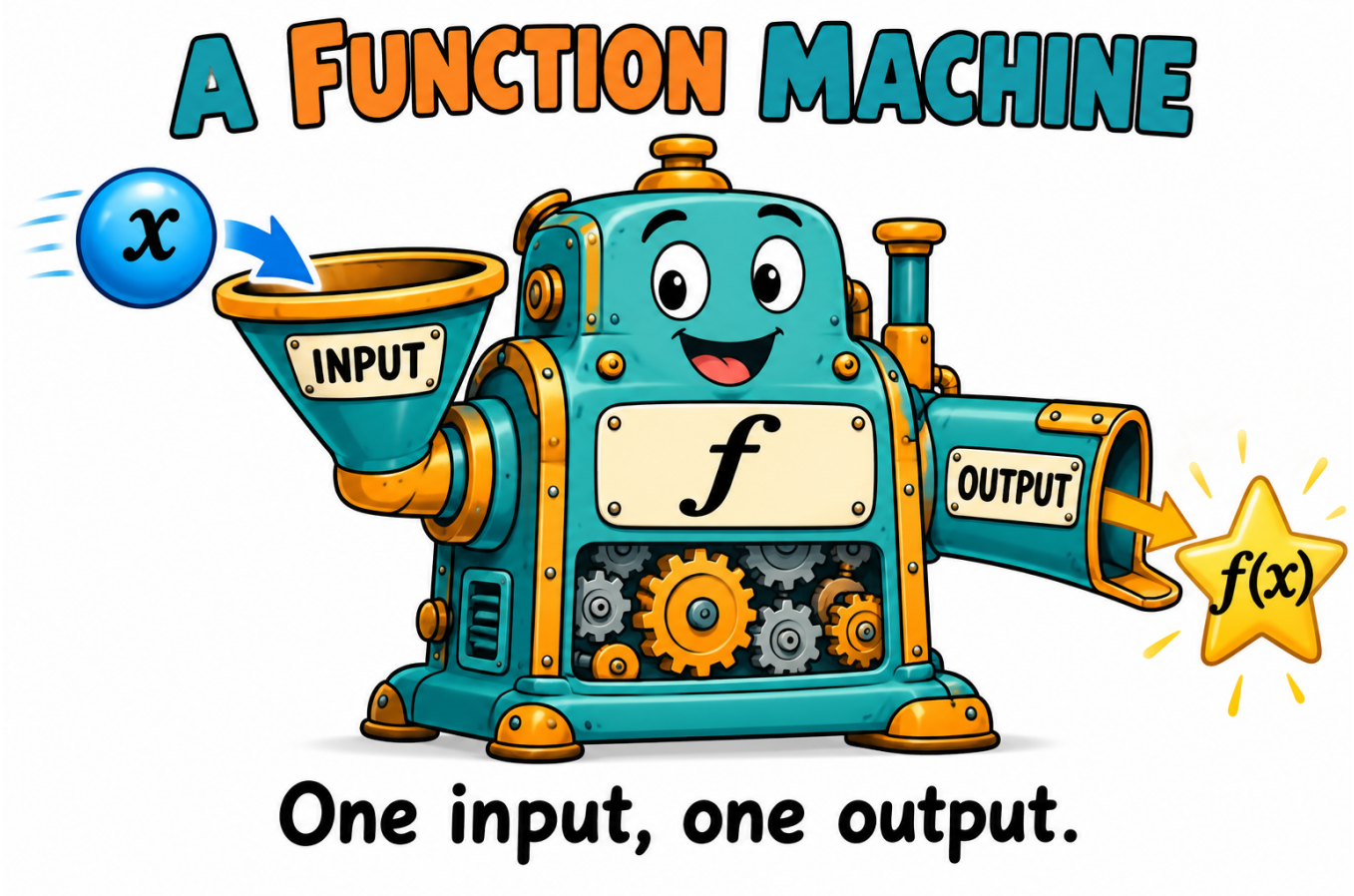
Intuition



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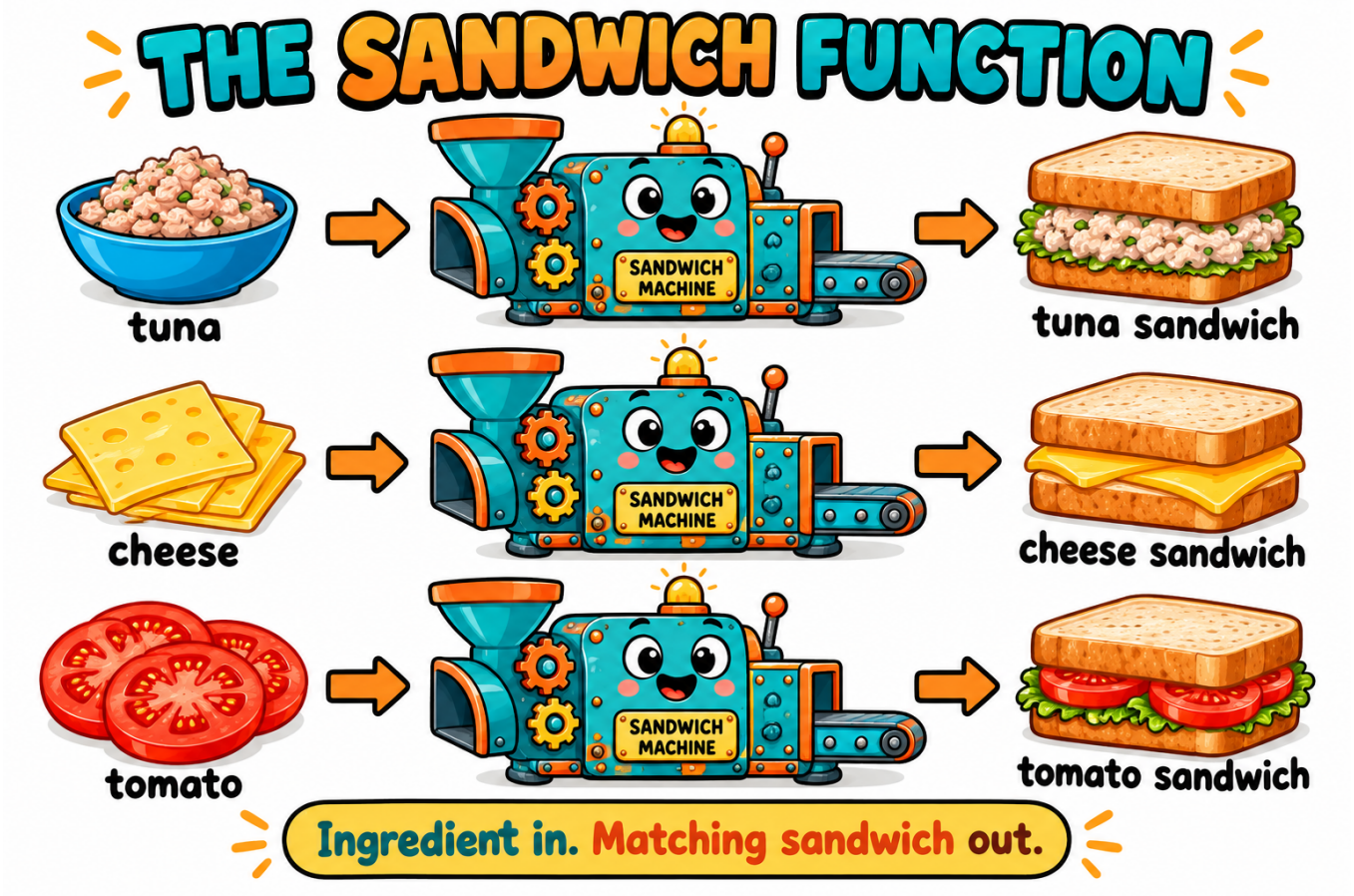
THE SANDWICH FUNCTION
SANDWICH MACHINE
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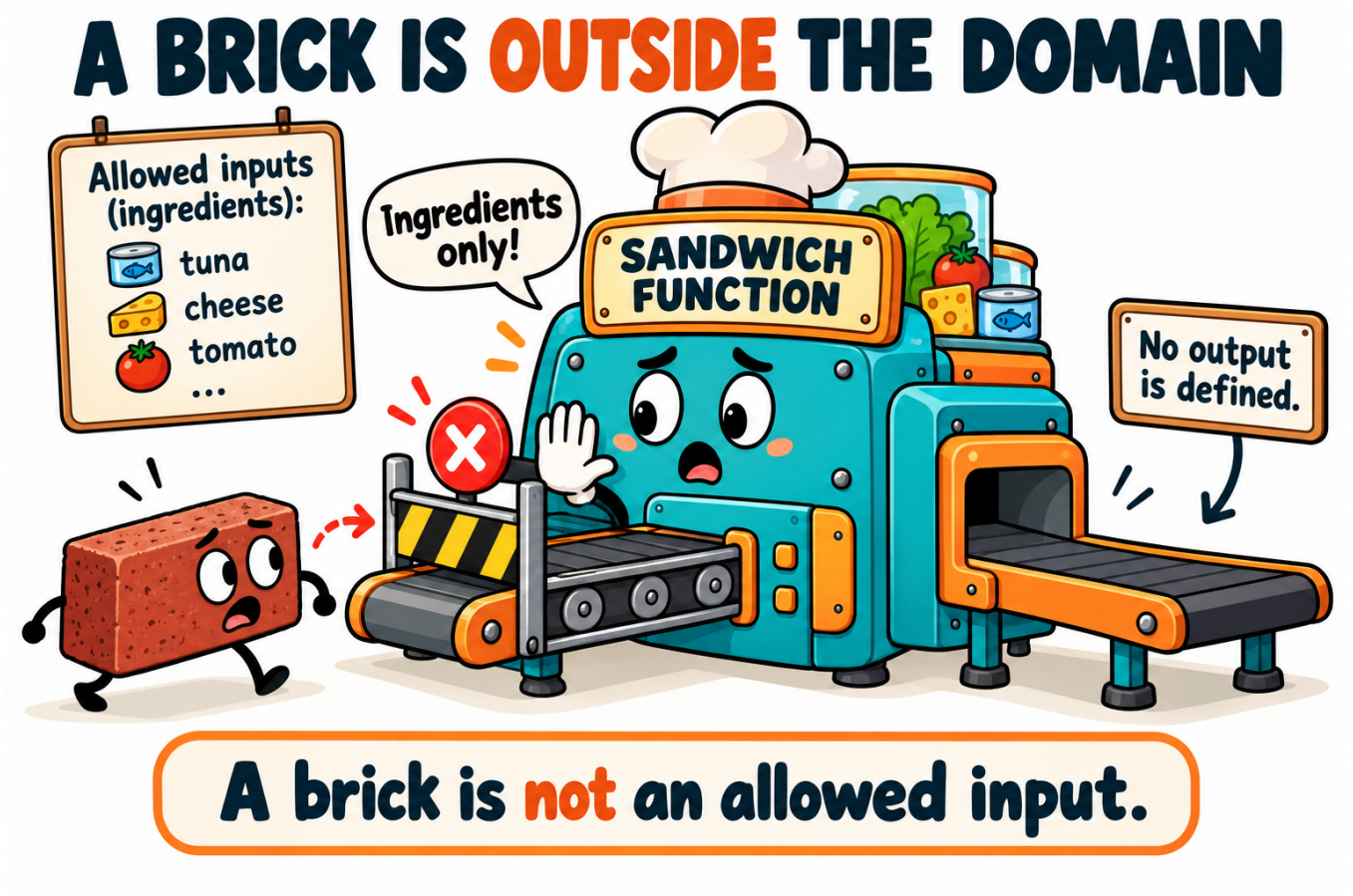
tuna
cheese
tomato
tuna sandwich
cheese sandwich
tomato sandwich
Ingredient in. Matching sandwich out.

Intuition


A BRICK IS OUTSIDE THE DOMAIN
Allowed inputs (ingredients):
tuna
cheese
tomato
...
Ingredients only!
SANDWICH FUNCTION
No output is defined.
A brick is not an allowed input.

Relations

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Definition

A **relation** $\mathcal{R}$ from A to B is a subset of the Cartesian product $A \times B$.

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The **domain** and **range** of $\mathcal{R}$ are

$$\text{Dom}(\mathcal{R}) = \{a \in A : \exists b \in B,\ (a, b) \in \mathcal{R}\}$$

$$\text{Rng}(\mathcal{R}) = \{b \in B : \exists a \in A,\ (a, b) \in \mathcal{R}\}$$

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$$\mathcal{R} = \{(2, \heartsuit), (5, \diamondsuit), (5, \spadesuit), (7, \heartsuit)\}$$

is a relation from A to B for

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$$\text{Rng}(\mathcal{S}) = [-2, 2]$$

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$$(\{1\}, \{1, 2\}) \in \mathcal{T}, \quad (\varnothing, \{2, 3, 5\}) \in \mathcal{T}, \quad (\{2\}, \{1\}) \notin \mathcal{T}$$

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Notation:

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- redundant notation!

$$f(a) = b \quad afb \quad \text{and} \quad (a, b) \in f$$

all mean the same thing!

Functions

Example

The relation

$$f = \{(1, \heartsuit), (2, \spadesuit), (3, \diamondsuit)\}$$

is a function from $A = \{1, 2, 3\}$ to $B = \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}$.

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- we can write $f(1) = \heartsuit$ or $1 f \heartsuit$ or $(1, \heartsuit) \in f$

Functions

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- $1f\heartsuit$ and $1f\diamondsuit$
- $f(1) = \heartsuit$ and $f(1) = \diamondsuit$

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Solution

Just have to decide the value of $f(1)$, $f(2)$, and $f(3)$.

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Solution

Just have to decide the value of $f(1)$, $f(2)$, and $f(3)$.

$$4 \times 4 \times 4 = 64 \text{ possibilities.}$$

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How many subsets of $A \times B$?

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Solution

How many subsets of $A \times B$?
How many elements in $\mathcal{P}(A \times B)$?

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Find how many relations there are from $A = \{1, 2, 3\}$ to $B = \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}$.

Solution

How many subsets of $A \times B$?
How many elements in $\mathcal{P}(A \times B)$?

$$|\mathcal{P}(A \times B)| = 2^{|A \times B|} = 2^{3 \times 4} = 4096$$

Rules

Defining functions by rules

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Examples:

- $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$

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Often, we define a function by specifying a **rule** for what $f(x)$ sends a value x to.

Examples:

- $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$
- $f : \mathbb{Z} \to \mathbb{Q}$ defined by

$$f(x) = \begin{cases} \frac{2x+1}{x-2}, & x \neq 2 \\ 3, & x = 2 \end{cases}$$

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- **injective** or **one-to-one** if for all $a_1, a_2 \in A$ if $f(a_1) = f(a_2)$ then $a_1 = a_2$

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- **surjective** or **onto** if for all $b \in B$ there exists $a \in A$ with $f(a) = b$

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- **bijective** if its both injective and surjective

Properties of functions

These are statements about the range!

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- **injective** means each $b \in B$ is hit **at least one time**
- **bijective** means each $b \in B$ is hit **exactly one time**

Example

Problem

What properties does the following function have?

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2$$

Example

Problem

What properties does the following function have?

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2$$

Solution

Neither injective nor surjective

Example

Problem

What properties does the following function have?

$$f : \mathbb{R} \to [0, \infty), \quad f(x) = x^2$$

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Solution

Surjective but not injective

Example

Problem

What properties does the following function have?

$$f : (-\infty, 0] \to [0, \infty), \quad f(x) = x^2$$

Example

Problem

What properties does the following function have?

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Solution

Both injective and surjective. Bijective!

Functions

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Solution

Choose where 1 goes. Then choose somewhere different for 2. Then yet another different place for 3.
Possibilities are $4 \times 3 \times 2 = 24$.

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Challenge

Find how many surjective functions there are from $A = \{1, 2, 3\}$ to $B = \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}$.

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Solution

None!

Cardinality

Definition

We say two sets have the same **cardinality** if there exists a bijection $f : A \to B$

- for finite sets, bijection iff $|A| = |B|$
- for infinite sets, things get interesting

Hilbert's Infinite Hotel

Problem

David Hilbert owns a rather particular hotel with infinitely many rooms labeled $1, 2, 3, \ldots$.
One night, the hotel is completely booked when a guest shows up and asks for a room. Hilbert says "No problem!" and makes room for the new guest. How does he do it?

Hilbert's Infinite Hotel

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Mathematically, we've built a bijection!

$$f : \mathbb{N} \cup \{0\} \to \mathbb{N}$$

$$f(x) = x + 1$$

The new guest is 0. The old guests are in rooms 2 and up. The new guest is in room 1.

Hilbert's Infinite Hotel

Problem

Later the same night, a bus full of infinitely more guests show up. Hilbert makes room again! How does he do it?

Hilbert's Infinite Hotel

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Hilbert's Infinite Hotel

Mathematically, we've built another bijection!

$$f : \mathbb{Z} \to \mathbb{N}$$

$$f(x) = \begin{cases} 2x, & x > 0 \\ 2x + 1, & x \leq 0 \end{cases}$$

The old guests are in even rooms. The new guests are in odd rooms.

Hilbert's Infinite Hotel

Outline

1. SoPS Lecture 19
   - Functions
   - Cardinality
   - Wrapping up

Final thoughts

Reflection

Final thoughts

Reflection

- Describe the difference between a relation and a function
- Give an example of a relation which is not a function. Explain why it isn't.
- Give an example of a function which is surjective but not injective.

Submit your responses by **Tonight!**